

MRINAL SINGH

Software Development Engineer (SDE) | Full-Stack & Applied Systems
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EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Bhubaneswar, Odisha, India

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.30 / 10.00

Aug 2024 – May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Linear Algebra, Probability & Statistics.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL

Frontend & Backend: React.js, Vite, Node.js, Express.js, FastAPI, RESTful APIs, Tailwind CSS, HTML5/CSS3

Databases & Cloud: PostgreSQL, Supabase, Vercel, Render

Data Science & ML: Scikit-learn, SHAP, NumPy, Pandas, Data Preprocessing, Predictive Modeling

Developer Tools & Core: Git, GitHub, Postman, Linux/Bash, Cryptographic Hashing (SHA-256), OOP, System Design

EXPERIENCE

GirlScript Summer of Code (GSSoC)

Remote, India

Open Source Software Developer Intern

May 2025 – Aug 2025

- Engineered bug fixes, modular features, and UI/UX performance enhancements across diverse open-source repositories.
- Collaborated with maintainers and cross-functional teams in an agile environment, adhering to code review guidelines and git workflows.
- Authored well-documented PRs, optimized frontend component rendering, and integrated clean REST API endpoints.

Open Source Community (Hacktoberfest & GitHub)

Remote, Global

Open Source Software Contributor

Oct 2025 – Present

- Successfully authored and merged **6+ pull requests** across production developer tools, utility libraries, and web platforms.
- Engaged in peer code reviews, resolved codebase issues, and maintained clean version control and CI/CD standards.

PROJECTS

CaseBrief | Cyber Forensics & Fraud Evidence Reconstruction Platform

[GitHub](#)

Python, SHA-256 Hashing, Rule Engines, Regex/NLP Extraction, JSON Pipelines

2025

- Engineered an enterprise-grade cyber-forensics platform to ingest multi-modal fraud evidence (WhatsApp exports, SMS, bank statements, DNS traces) into unified chronological audit timelines.
- Architected a zero-hallucination deterministic parsing engine with cryptographic **SHA-256 chain-of-custody** tracking, ensuring court-admissible forensic integrity and tamper-proofing.
- Automated cross-evidence entity extraction, discrepancy detection, and investigative checklist generation, cutting forensic triage time by **70%**.

ByteCargo | Full-Stack Logistics & Bidding Marketplace

[Live Demo](#) | [GitHub](#)

React.js, Node.js, Express.js, PostgreSQL, Tailwind CSS, REST API, Vercel, Render

2025

- Developed a scalable two-sided logistics marketplace enabling merchants to create shipment auctions and transporters to submit competitive real-time bids with pricing and ETA tracking.
- Implemented secure JWT authentication, Role-Based Access Control (RBAC) for merchant/transporter dashboards, and protected REST API endpoints.
- Engineered responsive interfaces with modular state management and deployed microservices on **Vercel** and **Render**.

SkillOutcome | Predictive Career Engine & ML Placement API (SIH)

[GitHub](#)

Python, FastAPI, Scikit-learn, Feature Engineering, PostgreSQL, Data Pipelines

2025

- Designed an ML-driven placement prediction system with role-based skill-gap recommendation for Smart India Hackathon (SIH PS135).
- Constructed automated ETL pipelines across **13 normalized relational tables** with validation splits and probability calibration models.
- Exposed predictive inference models via production **FastAPI REST endpoints**, achieving sub-100ms response latencies.

ACHIEVEMENTS & CERTIFICATIONS

Algorithmic Problem Solving: Solved **200+ DSA problems** on LeetCode with a **76%+ acceptance rate** ([s_mrinal](#)), mastering graphs, dynamic programming, trees, and core data structures.

Smart India Hackathon (SIH): Selected institutional participant/finalist for AI/ML problem statement SIH26135.

Certifications: Responsive Web Design (freeCodeCamp) | Intro to Generative AI Studio (Google) | Interview Readiness (TCS iON)